

IMSE 785 Big Data Analytics

(Prerequisite: CIS 209, STAT 350 or 510)

Updated 5/10/2026

COURSE OBJECTIVES: An introduction to storage, retrieval, analyses, and display of data sets so large and complex that traditional data processing and analysis applications cannot readily be used. Topics include big data management, data architecture of hosting big data, big data retrieval languages, parallel computing methods, big data analytical methods, and data visualization.

TEXTBOOK AND/OR OTHER REQUIRED MATERIALS:

Required: Bengfort, B. and Kim, J. (2016) Data Analytics with Hadoop, 1st Ed. O'Reilly.
Leskovec, J., Rajaraman, A. and J. D. Ullman (2020) *Mining of Massive Datasets*, 3rd ed., Cambridge, NY. (Note that 2nd ed will also work.)
Chambers, B. and Zaharia, M. (2018) Spark the Definitive Guide. 1st Ed. O'Reilly

Optional: White, T. (2012) Hadoop: The Definitive Guide, 3rd. O'Reilly
Ryza, S., Laserson, U., Owen, S., and Wills, J. (2017). Advanced Analytics with Spark, 2nd ed., O'Reilly.
McKinney, W. (2013) Python for Data Analysis, O'Reilly.
Muller, A. C. and Guido, S. (2017) Introduction to Machine Learning with Python: A Guide for Data Scientist, 1st Ed., O'Reilly.

(Please note that **all O'Reilly books are available free through K-State library**)

INSTRUCTORS: Shing I Chang
OFFICE: Room: NA
E-mail: changs@ksu.edu Class site: Canvas
TA: Parviz Ghafariasl Ganjinehketab, parvizghafari@ksu.edu

OFFICE HOURS: By appointment via e-mail or Zoom
LOCATION AND TIME: Online Only

TOPICS:

Part I. Introduction

- What is big data? Hype? Or Here to stay?
- What are the difficulties encountered for handling big data?
- Strategy Planning for big data management
- Team of big data analytics

Part II. Big Data Storage, Manipulation and Retrieval (Data Wrangling)

- Alternatives for storing big data – Map-reduced, Hadoop, and
- Databases – SQL (Hive) and NoSQL (Hbase) vs Hadoop Distributed File System (HDFS)
- Pig, Hive, Hue, Impala, Hbase, Spark SQL for Data Manipulation and Retrieval
- Data Ingesting

Part III. Data Visualization (to be embedded in all topics)

- Communicating your analytic results
- Simple Graph Techniques
- Charting with time element
- Principles of Data Visualization
- Multivariate Data Presentation
- Dashboard Design

Part IV. Big Data Analytics (Machine Learning and BDA)

- Parallel Computing – Map Reduce programming paradigm
 - WordCount of Text Documents
 - Page Rank of Web Sites

- Spark for Interactive algorithms
 - MLib
 - Interactive Spark using PySpark
- Data Integration from Multiple Data Sources
 - Snowflake
- Finding Frequent Itemsets
- Associate Rule Mining
- Clustering (k-means)
- Prediction: Logistic Regression, Random Forest, Decision Tree
- Dimension Reduction
- Big Data and AI

GRADING POLICY: Exams: 30%
 Discussion, Homework: 20%
 Term projects: 50%

COMPUTER: You will be assigned a VM account through your TA. A reliable high-speed internet connection is essential. Details will be available when class period starts. **If you plan to run various assignments on your own equipment**, we recommend that you own a PC with Intel iCore 7 processor with at least 16 GB of RAM and 1TB of hard disk to install Cloudera Quickstart and Spark.

HW/EXAM POLICY: Late homework or project report (less than a week from the due date) will be graded based on half of the total points. Absent quiz or exam will receive no point. All exams and quizzes are administered through Canvas with time limits. No proctor is needed.

ACADEMIC ACCOMMODATIONS FOR DISABLED STUDENTS:

Any student with a disability who needs an accommodation or other assistance in this course should make an appointment to speak with your instructor as soon as possible.

ACADEMIC HONESTY: Plagiarism and cheating are serious offenses and may be punished by failure on the exam, paper or project; failure in the course; and/or expulsion from the university. For departmental policy please visit <http://www.imse.ksu.edu/ethics.html>. Fall semester 1999 marks the beginning of Kansas State University's undergraduate Honor System. Please refer to <http://www.ksu.edu/Honor> for more details. Special notes about using generative AI such as **Chat-GPT**, you are encouraged to use it to help you debug or initiate coding projects. But **you will violate honor code if it is used without source quotation, substantial modification, validation, and analysis for answering homework or project questions.**